Chi-square (χ^2) test of independence

BLOCK 3

Do two CATEGORICAL variables go together?

Choose the test: what type is X? what type is Y?

X (predictor / group)	Y (outcome)	MAIN QUESTION	TEST
Numeric	Numeric	Move together?	Correlation/regression
Categorical: 2 groups	Numeric	Means differ?	t-test
Categorical: 3+ groups	Numeric	Means differ?	ANOVA
Categorical: 2+ categories	**Categorical: 2+ categories**	**Related counts?**	**χ^2**

Chi-square is DIFFERENT: BOTH variables are categorical.

One p-value idea, across ALL the tests

H_0 (null hypothesis) = assume NO real relationship / difference as a starting point

$p < .05$ • REJECT H_0

The data would be unusual if H_0 were true. We have evidence AGAINST the “no pattern” idea.

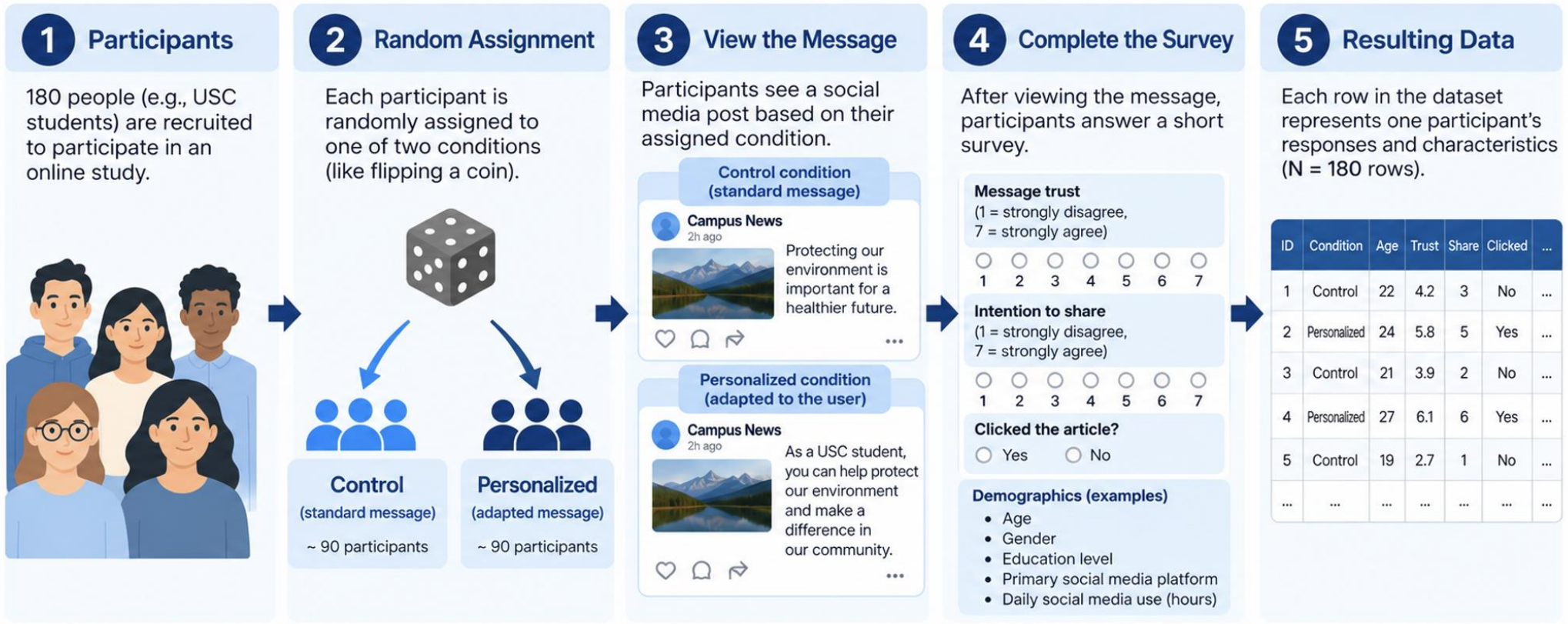
$p \geq .05$ • FAIL TO REJECT H_0

The evidence is not strong enough. This does NOT prove that H_0 is true.

$p < .05$ is a conventional cutoff—not a requirement that your data must satisfy.

Same dataset

A randomized experiment with two conditions and a post-message survey (N = 180)



Today we use this SAME dataset to ask three different statistical questions.

SAME dataset. Four different research questions.

r	**baseline_trust + message_trust**	Are two numeric scores associated?
t	**condition → message_trust**	Do two groups have different mean trust?
F	**education_level → message_trust**	Do three groups have different mean trust?
χ^2	**condition + clicked_article**	Are clicking rates related to condition?

The same participants can answer different questions when we select different columns.

Two columns from the dataset: labels, not scores

participant_id	condition	clicked_article
1	Control	**Yes**
2	Control	**No**
3	Control	**No**
4	Personalized	**No**
5	Personalized	**Yes**

Example rows 1–5 from the course CSV (real records, not the full dataset).

We tally each Yes / No inside each Control / Personalized group.

Chi-square gives TWO categorical variables a job

Research question: **Is article-clicking associated with message condition?**

X = message condition
Control / Personalized

↔

Y = clicked article
Yes / No

Both variables are CATEGORICAL → use a chi-square test of independence.

Chi-square asks about a RELATIONSHIP, not about average message trust.

First, make a two-way table of COUNTS

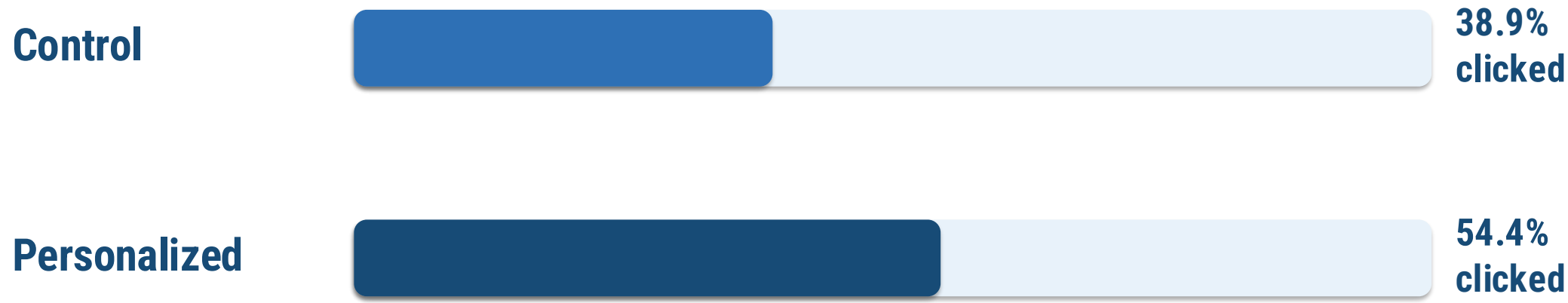
ILLUSTRATIVE COUNTS ONLY • Colab will display the real data

CONDITION	NO CLICK	CLICKED YES	TOTAL
Control	55	35	90
Personalized	41	49	90
Total	96	84	180

Counts tell us HOW MANY. We still need percentages to compare groups fairly.

Now compare ROW percentages, not just counts

ILLUSTRATIVE EXAMPLE (not the real CSV results)



35 / 90 = 38.9% clicked in Control

49 / 90 = 54.4% clicked in Personalized

The click rate helps us describe the direction and size of the sample pattern.

What is the chi-square NULL hypothesis?

H_0 • NULL

Message condition and clicking are INDEPENDENT.

The population click rate is the same in both conditions.

H_1 • ALTERNATIVE

Message condition and clicking are ASSOCIATED.

The population click rates are not the same.

The alternative says “related,” not necessarily which condition has a higher rate.

Expected counts: what would H_0 predict?

ILLUSTRATIVE 2 × 2 TABLE

OBSERVED (what we saw)

Control: 55 No • 35 Yes

Personalized: 41 No • 49 Yes

EXPECTED (if no association)

Control: 48 No • 42 Yes

Personalized: 48 No • 42 Yes

Expected = what we would anticipate under NO relationship between the categories.

How do we calculate one expected count?

Expected count = (row total × column total) ÷ grand total

ILLUSTRATIVE TABLE TOTALS

	Clicked Yes	Clicked No	Total
Control	?	?	90
Personalized	?	?	90
Total	84	96	180

We use the totals, not the observed cell counts, to compute what H_0 would predict.

WORKED EXAMPLE: CONTROL + CLICKED YES

Row total = 90 • Column total = 84 • Grand total = 180

(90 × 84) ÷ 180 = 42

So, if message condition and clicking are unrelated, **we would expect 42 Control participants to click Yes.** Repeat the same formula for each cell in the table.

Plain-language idea: expected counts = what we would expect to see if there were NO association.

What is χ^2 measuring? A mismatch score.

χ^2 looks at how different the OBSERVED counts are from the EXPECTED counts.

OBSERVED		EXPECTED
55	—	48
35	—	42
41	—	48
49	—	42

You do NOT need to memorize: $\chi^2 = \Sigma\ (\text{observed} - \text{expected})^2 / \text{expected}$

Bigger mismatches between observed and expected counts → larger χ^2, all else equal.

Read FOUR things from a chi-square result

01	**TABLE / %**	Which category combination is more common?
02	**χ^2**	How far are observed counts from expected counts?
03	**df, p**	df tells table complexity; p tests independence.
04	**CRAMÉR'S V (OPTIONAL)**	How strong is the association? (0 = none in sample)

Read proportions for meaning; read χ^2 and p for statistical evidence.

How to read one chi-square result

ILLUSTRATIVE NUMBERS ONLY — your Colab will show the actual results

$\chi^2(1) = 4.38$ $p = .036$ **Control: 38.9% Yes** **Personalized: 54.4% Yes**

1. % Personalized has a higher observed click rate in this sample.

2. p $p < .05$ gives evidence of association, assuming the test assumptions hold.

3. WRITE Message condition and clicking were associated in this illustration.

Significance says evidence of association, NOT that the difference is large or important.

What if the chi-square p-value is NOT below .05?

$p < .05$

Reject H_0.

The data give evidence that the categorical variables are associated.

$p \geq .05$

Fail to reject H_0.

The data do not give enough evidence of association.

"Not statistically significant" never means "proved no relationship."

Before trusting the result: quick checks

1 **ONE PERSON PER ROW**

No participant counted twice.

2 **COUNTS IN CATEGORIES**

Use frequency counts, not mean scores or percentages as input.

3 **EXPECTED CELLS**

For introductory Pearson χ^2, expected cell counts should generally be ≥ 5.

4 **DESIGN ≠ TEST**

Random assignment may support causal conclusions if other design assumptions hold.

The test does not establish causality by itself; experimental design matters.

Colab step 1: load and clean the SAME CSV


```
import pandas as pd

raw = pd.read_csv(url)
df = raw.drop_duplicates(
   "participant_id"
).copy()

df["condition"] = (df["condition"]
  .astype("string").str.strip().str.title())
```

WHAT THE NOTEBOOK DOES

- Reads the course CSV
- Keeps one record per person
- Standardizes “Personalized”
- Keeps Yes / No responses

A repeated ID or inconsistent spelling can create the WRONG category counts.

Colab step 2: make the contingency table


```
counts = pd.crosstab(
  chi_df["condition"],
  chi_df["clicked_article"]
)

display(counts)

row_pct = counts.div(
  counts.sum(axis=1), axis=0
) * 100
```

READ IT LIKE ENGLISH

- Rows = message groups
- Columns = click Yes / No
- Each cell = number of people
- Row % = within each condition

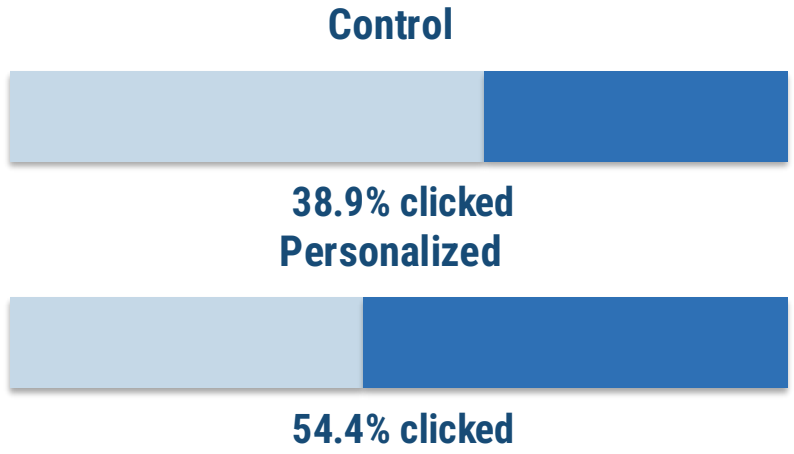
This is a COUNTS table: the input for chi-square.

Colab step 3: turn counts into a picture

COLAB LAB

```
row_pct.plot(
  kind="bar", stacked=True,
  figsize=(8, 4),
  color=["#C5D8E7", "#2E70B5"]
)

plt.ylabel("Participants (%)")
plt.xlabel("Condition")
plt.show()
```



Plot at right uses ILLUSTRATIVE percentages.

Compare proportions; raw counts can be misleading when group sizes differ.

Colab step 4: run the chi-square test


```
from scipy.stats import chi2_contingency

chi2, p, dfree, expected = (
    chi2_contingency(
        counts, correction=False
    )
)

print("Chi-square:", round(chi2, 3))
print("df:", dfree)
print("p:", round(p, 4))
```

WHAT TO LOOK FOR

- χ^2 measures mismatch
- df = degrees of freedom
- p decides whether to reject H_0
- Expected counts need a check

Read χ^2, df, p, and the EXPECTED-count table.

Colab step 5: what do the actual results mean?

1 COUNTS

How many people clicked in each condition?

2 PERCENTAGES

Which group has higher click rate, and by how much?

3 P-VALUE

Do we reject H_0 or fail to reject H_0?

4 SENTENCE

Is there evidence of an association between condition and clicking?

Your answer must describe the real output—not the slide's illustrative example.

Checkpoint: which analysis should YOU choose?

CHECKPOINT

Q1 Does message trust differ by 3 education groups?

Q2 Is clicking related to message condition?

Q3 Do baseline trust and message trust move together?

Q4 Does a baseline trust score predict message trust?

Ask what TYPE of outcome is being studied before selecting a test.

Exit ticket: explain chi-square in plain language

CHECKPOINT

- What TWO categorical columns did we compare?
- What does a contingency table count?
- Why do we calculate row percentages?
- What does a chi-square p-value < .05 mean?
- What must you check before making a causal claim?

If you can explain the question, the variables, and the p-value, you understand the logic.